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Rotary electroadhesive clutch

Abstract

An electroadhesive clutch using a ceramic-based dielectric layer separating opposing clutch plates, which function as electrodes. At least one of the clutch plates can be a flexible material, such as a thin film. A voltage applied across the electrodes creates an electrostatic attraction between the electrodes. The ceramic-based dielectric layer can be applied to one or both electrodes or, alternatively, placed between opposing electrodes. The ceramic-based dielectric reduces the voltage required to adhere adjacent plates, while also improving the force of adhesion. A rotary electroadhesive clutch using the ceramic-based dielectric provides improved force of adhesion and can be used to control the amount of torque transferred from an input to an output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 63/182,681, filed Apr. 30, 2021, which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The invention relates generally to an electroadhesive clutch, and more particularly to an electroadhesive clutch that uses a ceramic-based dielectric to separate adjacent clutch plates, and to a rotary clutch design employing a plurality of clutch plates with at least one set of flexible plates.

(2) Electroadhesive clutches use electrically conductive clutch plates that are separated by a dielectric material. When a voltage is applied across opposing clutch plates, where the plates are acting as electrodes, an electrostatic charge develops between the plates, creating an attractive state and causing the plates to adhere. With the plates adhered to each other, a force can be transmitted from one plate to the other. For example, a torque applied to one plate will transmit the torque to the opposing plate. Electroadhesive clutches can be created in various shapes, including rotary, stacked rotary, and linear, among others.

(3) While existing electroadhesive clutches demonstrate the ability to transmit forces, the amount of force transmitted is limited by the force of adhesion between the plates. The dielectric material used to separate the plates affects the force of adhesion in addition to the responsiveness of the clutch and residual adhesion or hysteresis. Prior clutches use polymer or ceramic particle-embedded polymer dielectric materials, which exhibit high hysteresis or require significant voltages for operation. In addition, polymer dielectric materials are susceptible to defects, such as voids, incomplete coatings, or inconsistent thickness. Particle-embedded polymers can suffer from agglomeration, poor mixing, or deterioration from aging. Other electroadhesive clutches employ rigid plates, which limit the surface contact between opposing clutch plates. It would therefore be advantageous to develop an improved dielectric material for electroadhesive clutches, improved methods for applying the dielectric material to the clutch plates, and clutch configurations that

enable greater forces of adhesion at lower voltages.

(4) Electroadhesive clutches are highly desirable for applications in which rotary motion needs to be controlled because of their low weight and low power consumption relative to other clutch technologies. In many applications, the required output torque exceeds the performance of a single pair of electroadhesive clutch plates given the practical constraints on the diameter of the device when it needs to fit into an assembly. This necessitates the use of multiple pairs of plates arranged in parallel such that their torque adds. However, using entirely rigid clutch plates in parallel presents many challenges, including issues with alignment, off-state friction, and effective load-sharing between pairs. It would be beneficial for a clutch configuration to overcome these challenges without requiring complicated alignment mechanisms or excessive plate pre-compression.

BRIEF SUMMARY

(5) Disclosed herein is an electroadhesive clutch using a ceramic-based dielectric layer separating opposing clutch plates. At least one electrode is coated with the ceramic dielectric layer, although both may be coated. The dielectric may also be disposed between adjacent plates, but not permanently adhered to either. The dielectric layer may comprise aluminum oxide, titanium dioxide, and other ceramic oxides, which can be applied by a variety of methods. These methods include dip coating, electroplating, anodizing, etching, sol-gel reaction, plasma electrolytic oxidation (PEO), plasma conversion, chemical vapor deposition, physical vapor deposition, sputtering, spin-coating, laser conversion or other surface chemical reaction. The improved ceramic dielectric reduces the voltage required to adhere adjacent plates, while also improving the force of adhesion. As opposed to polymer-based dielectrics, the ceramic-based dielectric material of the present disclosure can have fewer defect sites that may be liable to short circuit, incur a dielectric breakdown, or cause corrosion. The improved dielectric enhances clutch performance by reducing residual adhesion, offstate friction, and hysteresis.

(6) Further disclosed herein is a rotary electroadhesive clutch. The rotary clutch may comprise a pair of opposing clutch plates or a multitude of plates stacked in a unit. In an embodiment utilizing multiple plates, a first set of plates can be engaged with a shaft connected to the center of the plates and a second set of plates can be engaged with a housing connected to the outer periphery of the plates. A rotational force from the shaft will be transferred to the housing when the first set and second set of plates are engaged in an adhered stated.

(7) Further disclosed herein is a rotary electroadhesive clutch utilizing a plurality of clutch pairs with at least one flexible clutch plate in each pair. The rotary clutch may comprise a pair of opposing clutch plates or a multitude of plates stacked in a unit. In an embodiment utilizing multiple plates, a first set of plates can be engaged with a shaft connected to the center of the plates and a second set of plates can be engaged with a housing connected to the outer periphery of the plates. A rotational force from the shaft will be transferred to the housing when the first set and second set of plates are engaged in an adhered stated. The flexible plate enables the stacked configuration, as plate alignment is more easily enabled, clutch torque is increased, and off-state friction is reduced. In one embodiment, the flexible clutch plate is mounted to a rigid backing through selective attachment points to facilitate torque transmission through the structure of the clutch.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is an example of a linear clutch using a ceramic-based dielectric.

(2) FIGS. 2A-2B depict rotary clutches with stacked plates.

(3) FIGS. 3A-3B are rotary clutches with nested, stacked plates that transmit torque to a housing.

- (4) FIG. 4A-4B show a rotary clutch with concentric clutch plates.
- (5) FIG. 5 is a clutch used in an energy recycling actuator.
- (6) FIG. 6 is a rotary clutch with separate cylindrical surfaces.
- (7) FIG. 7 is a tube-shaped clutch.
- (8) FIG. 8 shows a combination linear and rotary clutch.
- (9) FIG. 9 shows a number of potential electrode compositions.
- (10) FIG. 10 shows a number of potential electrode configurations for a nested rotary clutch.
- (11) FIG. 11 shows a number of potential electrode configurations for a stacked rotary clutch.
- (12) FIG. 12 shows a number of patterning/texturing strategies.
- (13) FIG. 13 is a rotary clutch with multiple stacked clutch pairs with inner frames on ends of stack.
- (14) FIG. 14 is a rotary clutch with multiple stacked clutch pairs with inner and outer frames on ends.
- (15) FIG. 15 is a rotary clutch with multiple nested clutch pairs acting in parallel.
- (16) FIGS. 16A-16B show a frame aperture that allows passage of an electrode from one side to the other.
- (17) FIGS. 17A-17B demonstrate two different methods of tensioning the frame stack.
- (18) FIGS. 18A-18C depict electrical connections for the electrodes.
- (19) FIGS. 19A-19B are examples of rotary clutch configurations.
- (20) FIG. 20 is a flowchart showing a method of forming a ceramic dielectric layer.
- (21) FIG. 21 is a graphical flowchart showing a method of treating the surface of a dielectric layer.

DETAILED DESCRIPTION

(22) According to one embodiment of the disclosure is an electroadhesive clutch **100** achieving enhanced performance using a ceramic as the dielectric insulating material **101** between two adjacent electrodes, or clutch plates, **102**. In a neutral state, a thin air gap or low friction state exists between the adjacent, opposing electrodes **102**, allowing each electrode **102** to move independently. For example, with a rotary clutch **100**, one electrode **102** may rotate while the other remains stationary. The air gap can be up to a few millimeters or, alternatively, the electrodes **102** have minimal contact resulting in a low frictional force. When a voltage difference is applied across the two electrodes **102**, electrical charge is induced, or the electrodes **102** are otherwise charged or polarized, the separation between the two electrodes **102** is eliminated and they adhere to one another with a high attraction force, transmitting force across the interface when loaded. The voltage causes a positive charge to develop on one electrode **102** and a negative charge to develop on the opposing electrode **102**, developing an electrostatic adhesion. The force of adhesion will remain provided the two electrodes **102** do not directly contact each other with low electrical resistance, which would equalize the electric charges. The force of adhesion will initially remain if the power supply is disconnected without removing the voltage difference, but will dissipate over time due to leakage current. The force of adhesion may also be lost if the electrodes **102** are grounded to each other or there is a dielectric breakdown. The dielectric material **101** electrically insulates one electrode **102** from an opposing electrode **102**, preventing charge equalization.

(23) While the previous example clutch **100** has been described as having the opposing electrodes **102** in one of two states—fully engaged or disengaged—the clutch **100** can be operated in a partially engaged state. For example, FIGS. 2A-2B depict a rotary electroadhesive clutch **100** where the opposing electrodes **102** are controlled to permit limited rotational movement when engaged. Stated differently, in the active, partially-adhered state, one revolution in one electrode **102** does not result in a full revolution of the other electrode **102** because the surfaces of each electrode **102** are slipping relative to the other. This occurs when the voltage applied is insufficient to generate enough adhesion to fully transmit the applied torsional load. The torque threshold that separates the fully engaged state from the partially engaged state can be actively controlled in several ways: engaging less than the entire quantity of clutch plate **102** pairs, controlling voltage to modulate the

adhesion experienced, or other methods.

(24) In one embodiment, only one electrode **102** is coated with a dielectric layer **101**, but both electrodes **102** may be coated. For example, depending on the intended application, coating two electrodes **102** can increase wear life or reduce leakage current. The dielectric coating will also affect the coefficient of friction, which is necessary for producing a force. In an alternative embodiment, the dielectric material **101** is not adhered to either plate **102**. Rather, the dielectric material **101** is placed between adjacent plates **102**, such as a free-standing ceramic wafer or thin film with a ceramic coating.

(25) When the voltage is removed, the electrostatic attraction between the electrodes **102** releases. As shown in FIG. **1**, the clutch **100** includes at least one electrode **102** comprising a thin, flexible material. The thin, flexible material can include a foil, a polymer having a metal layer (i.e. aluminum-sputtered BOPET), an inherently conductive polymer or composite, carbon fiber or graphene, or similar electrically-conductive materials. The thin, flexible electrode **102** is able to deform, eliminating the air gap and conforming to the surface of the opposing electrode **102** when a voltage is applied. This enables both low off state friction when deactivated, and greater true surface contact when activated, which enhances on-state clutch plate adhesion and torque transmission. In some embodiments, the thin electrode **102** can be attached to a rigid substrate **110** to enable transmission of a force, while still allowing the thin electrode **102** to deform during operation. For example, FIG. **2A** shows a flexible electrode **102** connected to a rigid frame assembly **110** through the use of a pressure sensitive adhesive, selective sintering, welding, rivets, swages, bolts, heat stakes, stitches, adhesives, or epoxies **105**. If the opposing electrode **102** is thick and/or rigid, a force can be transmitted without the use of a substrate **110**.

(26) Higher performance and lower operating voltage are enabled by the use of a ceramic-based dielectric insulating layer **101**, compared to previously demonstrated polymer insulators and composite insulators made of ceramic particle-embedded polymers. Table 1 shows a comparison of some common dielectric materials and the ceramic-based dielectric materials of the present disclosure. The force per unit area of the clutch **100** is dependent on the thickness of the dielectric insulating layer **101** separating the electrodes **102**, the voltage applied, the dielectric constant of the dielectric material **101**, breakdown strength, and surface resistance of the dielectric insulating layer **101**, and the ability of the overall clutch plate structures **102** to conform and allow good surface contact at the clutch interface.

(27) The force/voltage hysteresis refers to unwanted residual adhesion and voltage that remains even after the voltage is removed and can reduce clutch **100** responsiveness or holding force on subsequent charge and discharge cycles. This can result from charges becoming trapped in the surface of the dielectric material **101**, or from the dielectric material **101** itself becoming semi-permanently polarized. The magnitude of both of these effects and their impact on clutch hysteresis are dependent on the characteristics of the dielectric material **101** utilized in the clutch **100**. This force/voltage hysteresis is further affected by clutch symmetry or non-symmetry, which refers to configurations where both clutch electrodes **102** are coated with the dielectric material **101** (i.e. symmetric) or only one electrode **102** is coated (i.e. non-symmetric). The clutch **100** performance can also be affected by how the charge (i.e. negative or positive) is applied to the coated electrode **102** in the non-asymmetrical clutch **100**. Ceramic-based dielectric materials **101** show lower clutch hysteresis behavior.

(28) TABLE-US-00001 TABLE 1 Dielectric layer Force per Force/ Dielectric thickness unit area Operating Voltage material (um) (N/cm²) voltage (V) Hysteresis Commonly Polyimide, 5-50 0.1-2 500-2000 High used polymer polyethylene, dielectric PVDF materials Prior art DUPONT 20-100 1-2 250-350 Low dielectric Luxprint 8153 layers (years ceramic 2014-2021) particle- embedded polymer Aluminum 0.5-50.sup. 4-30 12-300 Very Low Oxide Titanium 1-10 20-100 5-200 Very Low Dioxide

(29) With a lower dielectric layer thickness, lower voltages may be used to achieve high holding

forces. However, with thinner dielectric layers, there is also a greater risk of electrical shorting occurring at defect sites, common in polymer-based coatings. Ceramic materials can be applied as thin, defect-free layers by chemical, vapor deposition, or electrochemical means, reducing the risk of electrical shorting.

(30) The ceramic dielectric layers **101** of the clutch **100** can be created by anodizing a metal substrate, such as aluminum, titanium, magnesium, zinc, zirconium, tantalum, and other metals, to create the ceramic layer directly on the metal, which acts as the electrode **102**. The clutch plate **102** can be formed from the metals or alloys containing the metals. Alternatively, the clutch plate **102** may comprise a substrate **110** with a metal or metal alloy disposed on the surface of the substrate **110**. For example, the dielectric layer **101** can be created by sputtering metals onto a substrate and processing the metal through an oxygen chamber to oxidize the surface of the thin metal while leaving some of the metal layer intact beneath the oxidized layer, in order to act as the conductive electrode **102**. In this case, the substrate **110** acts as a carrier to facilitate production, and to help transmit force through the clutch plate **102** under loading during clutch **100** operation. Another method includes chemical vapor deposition or sputtering to deposit ceramic directly onto a surface on an electrode **102**. In another embodiment, a free-standing ceramic dielectric layer **101** can be sputtered with a metal layer on one side to create a clutch plate **102** where the dielectric layer **101** will contact the opposing clutch plate **102** surface.

(31) The ceramic layer **101** can also be based on a single or multiple types of metal cations in combination with common anions including carbide, oxide, nitride, sulfide, fluoride, silicate, titanate, zirconate, and aluminate. In these examples, nano- or microparticles of ceramic can be embedded into a polymer matrix to form a paintable, printable, or sprayable material. Other solvents or carriers can also be used to enable application of the dielectric material **101** onto the surface of the clutch plate **102**. The particles can have a size of 1-100 μm , or alternatively, 1-1,000 nm. In one embodiment, the dielectric material comprises barium titanate (BaTiO_3) dispersed in a fluoropolymer matrix.

(32) In the methods described above, the dielectric layer **101** is formed of a substantially homogeneous or uniform layer of ceramic typically with at least 50% weight percentage. The dielectric layer **101** may have imperfections and/or a microstructure that results from the anodization process or other processes. Further, the dielectric layer **101** may include small quantities of additives (for example, pigments for color). In one embodiment, the dielectric layer **101** is a uniform oxide ceramic, forming a continuous layer on the surface of the electrode **102**.

(33) In the example of anodizing, a metal surface is immersed in a highly acidic solution and used as an anode to complete an electrochemical oxidation process, as is known in the art. This is most commonly applied to aluminum alloys but can also be applied to similar metals, titanium, zirconium, niobium, zinc, tantalum and magnesium. Many acidic solutions can be used as the electrolyte, most commonly including sulfuric acid, boric acid or chromic acid. The process is generally applied between 1 and 100V and controlled by cooling the solution, time of application and voltage. The resulting surface is generally porous, with columnar pores on the order of 1-100 nm diameter which may be dyed using common dye chemicals and sealed using boiling water or chemical sealing agents to give a more robust surface. Other examples of oxidative processes can be used to give a pore-free surface instead, for example plasma electrolytic oxidation. This process can be controlled more precisely to achieve consistent dielectric layer **101** thickness, uniformity, and low defects, compared to screenprinted composite dielectrics, for example.

(34) After anodizing, the dielectric **101** may have a natural columnar porosity. A heat treatment can be used to remove pore bound water, changing the overall dielectric properties of the layer **101** and to limit Faradaic processes under voltage application. For example, Faradaic currents may result from water-based electrochemical reactions. Excluding water from the dielectric layer **101** can reduce these reactions. Liquids, gases, or oils can also be introduced into the pores of the anodized dielectric layer **101** through immersion, sonication, vacuum, spraying, and other similar techniques.

(35) Liquids may include water miscible lab solvents, such as ethanol, methanol, isopropanol, acetone, THF, acetonitrile, dioxane, and DMSO. Water immiscible solvents can also be used, such as hexane, toluene, benzene, butanol, dichloroethane, dichloromethane, MEK, chloroform, ethers, aliphatic alcohols, aromatic alcohols, xylene. These water immiscible solvents have the ability to improve dielectric properties and to exclude water from the surface of the dielectric layer **101**. Dielectric oils can also be introduced into the pores of the anodized layer **101**. Oils may include mineral oils, silicone oils, vegetable oils, petroleum oils, and other similar oils. Gases introduced into the pores of the anodized layer **101** include inert gases, such as nitrogen, argon, helium, krypton, air, carbon dioxide, sulfur hexafluoride, carbon monoxide, nitrous oxide.

(36) The surface morphology of the dielectric layer **101** can be controlled through the application of heat and/or pressure using a flat press, roller, or similar device. The press, roller, and other devices can also be used to impart a texture or pattern to the surface of the dielectric **101**, which affects the coefficient of friction and holding force in the electroadhesive clutch **100**. Textures and other surface features **120** may include nano- or micro-scale holes, trenches, pores, embossed regions, debossed regions, pillars, waves, ridges, dimples, zigzags, slits, dents, or selective non-conductive coatings. Textures and features **120** may also be imparted using blades, laser ablation, patterned masks, media blasting, selective coating, chemical etching, and electrochemical dissolution. The surface of the dielectric layer **101** and/or electrodes **102** may have selective roughness, a mixture of textures and features **120**, or regions with features/roughness combined with smooth regions. Surfactants such as silanes or siloxanes can also be used to modify the surface of the dielectric layer **101** and/or electrodes **102**.

(37) The devices **100** described herein may be used as clutches, brakes, dampers, or torque limiters and can be used to prevent relative motion between two components. Multiple uses of each device **100** is enabled by strategic control of the applied voltage. High voltages will enable the devices to produce large forces or torques to resist motion or lock the relative position of components. Medium voltages will supply lesser forces or torques which may be overcome by the user or driving actuator. In this case, the devices **100** described here act as torque transmitters, dampers or resistive mechanical loads.

(38) The figures show various clutches **100** utilizing the solid ceramic-based dielectric layer **101**. FIG. 1 shows a linear clutch **100** with one thin, flexible electrode **102** and one thick electrode **102**, both coated with a dielectric material **101**, where the thin electrode **102** is loaded in tension, and the thick electrode **102** used to transmit an in-plane force in multiple directions or an in-plane torque. Alternatively, the linear clutch **100** may comprise two thin flexible electrodes **102**, one or both coated with a dielectric layer **101**. The electrodes **102** transmit force through tension.

(39) FIGS. 2A-2B show a linear clutch **100** with one thin electrode **102** selectively adhered to a rigid substrate **110** and one thick electrode **102**, one or both coated, or two thin electrodes each adhered to a rigid substrate **110** where the clutch **100** can be loaded in in-plane tension, in-plane compression, or in-plane torsion.

(40) FIGS. 3A-3B depict a rotary clutch **100** with parallel plates **102** that transmit torque between an input shaft **130** and a housing **131**, where the plates **102** are oriented perpendicular to the shaft **130**. A thick electrode **102** can be used for both electroadhesion and force transmission, while a thin electrode **102** is connected to a frame or substrate **110** that transmits torque between the electrode **102** and the housing **131** or shaft **130**.

(41) FIGS. 4A-4B show a rotary clutch **100** consisting of a tube-shaped thick electrode **102** or curved thick electrode **102**, and a thin electrode **102** draped around the circumference of the thick electrode **102** and connected to a torque-transmitting frame **110**.

(42) FIG. 5 is an energy recycling actuator using multiple double-clutched springs in parallel with one another, where the connection of each spring to either the output or the housing is controlled with electroadhesive clutches **100**.

(43) FIG. 6 shows a rotary clutch **100** in which two separate cylindrical surfaces (concentric and of

equal diameter) are wrapped by a thin electrode **102** that bridges the gap between the cylindrical surfaces, acting as a second electrode **102**. The cylindrical surfaces **102** may be ceramic coated, forming the dielectric layer **101**. When the clutch **100** is active, the thin electrode **102** adheres to both cylindrical surfaces **102** thus locking them together. When the clutch **100** is disengaged, the two cylindrical surfaces **102** can rotate freely relative to one another. The portion of the thin electrode **102** that bridges the gap between cylindrical components may be thicker to prevent wrinkling or reinforced by adhering it to a strip of stronger or stiffer material. The collar may be made out of rolled sheet metal that is bent to a diameter slightly smaller than the diameter of the cylindrical component **102** such that it hugs the cylinders **102** when it is stretched and placed over the cylinders **102**. Alternatively, the thin electrode **102** can be permanently attached to one of the two cylindrical surfaces **102** while allowing the thin electrode **102** and other cylindrical surface to interact in the same way.

(44) By way of further detail, as shown in FIG. **6**, the two cylindrical surfaces **102** share a common axis of rotation (left image); a thin, flexible rolled collar is present (middle image); and the collar is wrapped around both cylinders **102** such that the gap between them is bridged. The collar may be permanently attached to one or neither of the cylinders **102**. When voltage is applied to the collar and the cylindrical surfaces **102** they adhere, thus locking the relative angular positions of the cylinders **102** together. The dielectric coating **101** may be applied to the collar, the cylinders **102**, or both.

(45) FIG. **7** is a linear clutch **100** in which one electrode **102** is tube shaped and the other electrode **102** consists of a brush attached to a Bowden cable. The undeformed diameter of the brush is wider than the inner diameter of the tube such that the brush deforms and maintains contact with the sides of the tube when it is inserted. The brush bristles are thin and wide and may be made of metallic components or carbon fiber. When charged, the bristles adhere to the lining of the tube. One or both the bristles and the tube's inner liner may be coated with a dielectric material **101**.

(46) As shown in FIG. **7**, this clutch **100** is to be used inside a hollow tube that may be flexible or rigid. A cable or other long thin element runs through the hollow tube and moves relative to the tube along the tube's axis. The brush bristles are composed of thin sheets of conductive material that conform to the inner shape of the tube and maintain contact with the inside surface of the tube. The inside of the tube is coated with a dielectric and the brush may be bare conductive material or coated as well. The bristles of the brush adhere to the inner surface of the tube when voltage is applied. The bristles may be incorporated into the cable during the cable twisting process.

(47) FIG. **8** is a cross sectional view of a combination linear and rotary clutch **100**. One electrode **102** is tube shaped and the other electrode **102** is cylindrical and fits inside the tube. The inner cylindrical component is free to both rotate and translate around (i.e. rotate) and along (i.e. telescope) the axis of the tube. Contact is maintained between the cylinder and the tube either through a brush or via a low stiffness gap filling material that is coated with conductive material. If the gap filling material is conductive itself, an additional coating will not be necessary.

(48) The clutch is split into two components: the outer housing (white) and the inner shaft (shades of grey). The outer housing is hollow with an internal cylindrical surface that is coated with a dielectric material **101**. The inner shaft is fixed to a compressible gap-filling material such as a rubber gasket or foam. This gap filling material is coated with a conductive material. When voltage is not applied, the shaft is free to slide along and rotate about the housing's axis. When voltage is applied, the conductive coating on the gap-filling material adheres to the housing's inner surface. This locks the joint preventing both relative rotation and sliding.

(49) FIG. **9** shows a number of potential electrode **102** configurations. In order for the clutch **102** to operate, each clutch pair must include two electrodes **102** separated by a dielectric material **101**. The dielectric **101** may be on one electrode **102** or both. Each individual electrode **102** may be a bare metal or other conductive material, or a conductive layer deposited on or adhered to a carrier **110** that may be insulating or conductive. In some cases, the electrode may include a conductive

electrode on both sides of the carrier **110**. Each of these configuration is also possible with the addition of a dielectric coating **101**. In some embodiments, the carrier **110** with conductive electrodes **102** on both sides may have insulating dielectric **101** coated on the surfaces of both electrodes **102**.

(50) FIG. **10** shows a simplified crosssectional view of a number of potential electrode **102** configurations for a nested rotary clutch **100**. The basic configuration shows a rigid outer frame **136** in the shape of a ring (darkest gray) connected to two electrodes **102** (medium gray) with a double sided adhesive (speckled). The rigid, ring-shaped outer frame **136** and its electrodes **102** enclose an inner frame **135**. The outer **136** transmits mechanical load to the housing **131** and the inner frame **135** transmits mechanical loads to the shaft **130**. The inner and outer electrodes **102** can each be flexible or rigid in any combination other than both being rigid. Dielectric material **101** may be deposited on both electrodes **102**, just the inner electrode **102**, or just the outer electrode **102**. Nine configurations demonstrating combinations of these features are shown in FIG. **10**. These configurations are arranged in a grid. The top row represents all the configurations in which both the inner electrode **102** is rigid and acts as both the electrode **102** and the frame **135**. The middle row represents all the combinations in which the inner frame **135** includes a flexible electrode **102** and the outer electrodes **102** are rigid (black). The bottom row represents all the configurations in which both the outer and inner electrodes **102** are flexible. The left column represents all the configurations in which dielectric material **101** is deposited on both inner and outer electrodes **102**. The middle column represents all the configurations in which dielectric material **101** is deposited only on the outer electrode **102**. The right column represents all the configurations in which a dielectric coating **101** applied to the inner electrode **102** and not the outer electrode **102**.

(51) FIG. **11** shows a simplified crosssectional view of a number of potential electrode **102** configurations for a stacked rotary clutch **100**. The basic configuration shows a rigid outer frame **136** in the shape of a ring (darkest gray) connected to an electrode **120** (medium gray) with double sided adhesive applied in a pattern (speckled). The rigid, outer frame **136** is stacked adjacent to an inner frame **135** such that their electrodes **102** are touching or have a small airgap. The outer **136** transmits mechanical load to the housing **131** and the inner frame **135** transmits mechanical loads to the shaft **130**. The inner and outer electrodes **102** can each be flexible or rigid in any combination other than both being rigid. Dielectric material **101** may be deposited on both the outer and inner electrode **102**, just the inner electrode **102**, or just the outer electrode **102**. Nine configurations demonstrating combinations of these features are shown in FIG. **11**. These configurations are arranged in a grid. The top row represents all the configurations in which both the inner electrode **102** is rigid. The middle row represents all the combinations in which the inner frame **135** includes a flexible electrode **102** and the outer electrodes **102** are rigid (black). The bottom row represents all the configurations in which both the outer and inner electrodes **102** are flexible and both adhered to a rigid frame **135/136** using patterned adhesive. The left column represents all the configurations in which dielectric material **101** is deposited on both inner **135** and outer frames **136**. The middle column represents all the configurations in which dielectric material **101** is deposited only on the outer electrode **102**. The right column represents all the configurations in which a dielectric coating **101** is applied to the inner electrode **102** and not the outer electrode **102**.

(52) FIG. **12** shows a number of patterning and texturing strategies that can be used to apply features **120** to the electrodes **102** or dielectric layer **101** to provide improved release after deactivation and/or to strategically change the effective coefficient of friction to improve on-state holding force or reduce off-state friction. These strategies include (from left to right, top to bottom): holes, embossing, ridges, trenches, debossing, slits, pores, pillars, and regions of increased roughness. The locations of the features **120** shown are only illustrative. These features **120** can be of any size and can be located anywhere on the surface of the clutch plate **102** or dielectric **101**,

even non-symmetrically. Also, multiple types of features **101** may be combined on a single plate **102** or across two paired clutch electrodes **102** to create different patterns or effects.

(53) FIG. **13** is a rotary clutch **100** with multiple stacked clutch pairs with inner frames **135** on the ends of a stack of clutch plates **102**. Compression is maintained through the stack using a wave washer **138** on the top inner frame **135** and a retaining ring **137** mounted in a groove on the shaft **130** at the bottom of the stack. Any multitude of clutch frames **135/136** can be stacked on top of each other to increase the torque capacity of the clutch **100** as a whole. The shaft **130** is connected to the cylindrical housing **131** through two bearing connections in hubs on the face of the housing **131**. The inner frames **135** interact with the shaft **130** and the outer frames **136** interact with the housing **131**. This embodiment may also be altered to directly include a motor, gears, or other actuator or transmission component within the housing **131**. Alternatively, the clutch plates **102** may be placed without these auxiliary components into the housing of a gearbox, motor, or other actuator without the need for duplication of pre-existing auxiliary components in these devices such as bearings.

(54) FIG. **14** shows a stacked style clutch **100** in which the outermost clutch frames **136** are an outer on top and an inner frame **135** on the bottom. Compression between the frames **135/136** is maintained via a wave washer **138** that rests on the inner race of the bottom bearing. The top of the stack is supported by an internal retaining ring **137** located in a groove in the housing **131**. The shaft **130** is constrained by a retaining ring **137** that rests on the inner race of the top bearing.

(55) FIG. **15** shows a nested style clutch **100** in which the outer frames **136** are ring shaped and the inner frames **135** are enclosed within this ring **136** by the outer frame's electrodes **102**. In this illustration spacers **139** are placed between the nested frames **135/136** to manage spacing between the nested frames **135/136**, but spacers **139** are not always necessary for this design. Alignment is managed by a wave washer **138** that rests on the inner race of the bearing. The bottom of the stack of clutch plates **102** rests on the retaining ring **137**, which in turn rests on the other bearing. Again, the wave washer **138** and retaining ring **137** are not always necessary for this design because of the semi-independent nature of the nested clutch plate pairs **102**.

(56) By way of further detail, the clutches shown in FIGS. **13**, **14** and **15** may be modified in a number of ways including but not limited to the following: Any number of clutch frames **135/136** may be added or removed to achieve the desired torque capability, any retaining ring **137** may be replaced by a step, shoulder, or other mechanical restraint, ball bearings may be replaced by plane bearings, bushings, air bearings, or absent altogether, the housing **131** and hubs may be explicit components, or they may be features of a larger housing **131** that serves multiple functions in a greater product such as a robot.

(57) FIG. **16A** shows the exploded view of a configuration for an inner frame **135**. The rigid frame **135** includes teeth for interacting with a splined shaft **130**, and four apertures **140**. The circular apertures **140** are for inserting swages that can be used to clamp the flexible electrode **102** to the frame to establish electrical connections. The rectangular apertures **140** are for passing the electrode tails through to the opposite side of the frame **135**. This can be advantageous for avoiding electrical contact between inner **135** and outer frames **136** or for simplifying electrical connections. The flexible electrode **102** is then mechanically connected to the frame **135** via patterned adhesive. FIG. **16B** shows the same configuration from a cross-section view to better demonstrate the passing of the electrode tails through the aperture **140**. One, two or multiple tail passings may be used on a single frame **135**.

(58) FIGS. **17A-17B** show two methods for establishing compression along the clutch stack. In FIG. **17A**, the shaft **130** is threaded and a nut is placed on the threads and tightened to press the stack up against a retaining ring. A washer may or may not be placed on the retaining ring to better distribute the load. In the second configuration shown in FIG. **17B**, a shoulder is included on the shaft **130** and fasteners such as rivets or bolts are passed through the clutch stack **100** and fastened to the shoulder of the shaft **130**. Note, the clutch stack **100** may also be compressed in this way

without including the clutch shaft **130**. This creates a cassette of frames that can be inserted into the housing **131** or onto the shaft **130** as a collection rather than one at a time. Any combinations of fasteners, shoulders, retaining rings and grooves may be used to achieve the desired effect of compressing the frames **135/136** and maintaining position of the frames **135/136** relative to the shaft **130** axially.

(59) FIGS. **18A-18C** demonstrate three methods of establishing electrical connection to the clutch plates from the voltage supply. FIG. **18A** shows the clutch **100** with the housing hidden to demonstrate a configuration in which a spring finger **150** connects outer frames **136** to a command rail **151** mounted to the housing. FIG. **18B** shows an embodiment in which a wire is connected to the outer frames **136** through a solder pad or surface connector **152**. Teeth may or may not be selectively omitted from the frame **136** to allow space for the wired connection. FIG. **18C** shows an embodiment where the conductive nature of the frame **135/136** itself is utilized. There are no electrical connectors added to the frames **135/136**. Electrical connection is made between the inner clutch frame **135** to the shaft **130** via mechanical contact. Depending on the clutch **100** configuration, the top and bottom surfaces of the clutch frame **135** may or may not be conductive themselves or they may be coated with a dielectric material **101**. In a similar nature, the outer frames **136** may establish connection to electrically active housing **131** teeth.

(60) FIG. **19A** is a crosssectional view of an example clutch **100** configuration. The inner frames **135** are connected electrically via conductive spacers **139** or spring fingers **150**. The stack of inner frames **135** and spacers **139** is electrically connected to the voltage supply via a slip ring in which there is a rotating disk with slip ring brushes or spring fingers **150** attached. These brushes connect to a stationary portion of the slip ring that is mounted to the housing **131** via a sliding mechanical connection. FIG. **19B** shows an example of an inner clutch frame **135** with a conductive surface on the top and bottom faces for the conductive spacers **139** to contact. The remaining surface may be conductive or coated with a dielectric material **101** depending on the configuration. By way of further detail, the spacer **139** may be insulated on the inner diameter to prevent electrical connection from being established to the shaft **130**.

(61) FIG. **20** describes a flow chart for a sequence of potential modifications of an aluminum surface which is first anodized at step **201**, then oven dried to remove pore bound water at step **202**, further modified with a hydrophilic, hydrophobic or dielectric oil at step **203**, and then finally sealed at step **204**. The clutch **100** may be operative at any state of modification but lifetime, wear, leakage current or holding force can be improved by using combinations of these steps.

(62) FIG. **21** shows schematics of the modifications described in the FIG. **20** flow chart as an illustration of the anodized pores being modified by drying, liquid impregnation, and sealing to encapsulate a liquid. The surface may be used effectively for the clutch **100** as any of these stages effectively but performance may be improved for different embodiments of the clutch **100** by one specific modification over another.

(63) When used in this specification and claims, the terms “comprises” and “comprising” and variations thereof mean that the specified features, steps or integers are included. The terms are not to be interpreted to exclude the presence of other features, steps or components.

(64) The invention may also broadly consist in the parts, elements, steps, examples and/or features referred to or indicated in the specification individually or collectively in any and all combinations of two or more said parts, elements, steps, examples and/or features. In particular, one or more features in any of the embodiments described herein may be combined with one or more features from any other embodiment(s) described herein.

(65) Protection may be sought for any features disclosed in any one or more published documents referenced herein in combination with the present disclosure. Although certain example embodiments of the invention have been described, the scope of the appended claims is not intended to be limited solely to these embodiments. The claims are to be construed literally, purposively, and/or to encompass equivalents.

Claims

1. A rotary electroadhesive clutch comprising: a first set of clutch members, wherein the first set of clutch members are electrically conductive; a second set of clutch members interposed between the first set of clutch members, wherein the second set of clutch members are electrically conductive; and a dielectric material separating each clutch member of the first set of clutch members from an adjacent member of the second set of clutch members, wherein at least one of the first set of clutch members and the second set of clutch members comprises a flexible film, membrane, or sheet, in which the flexible film, membrane, or sheet is selectively or uniformly attached to a structural frame using pressure sensitive adhesive, selective sintering, welding, rivets, swages, bolts, heat stakes, stitches, adhesives, or epoxies.
2. The rotary electroadhesive clutch of claim 1, wherein the clutch members are circular and are aligned parallel to one another along an axis.
3. The rotary electroadhesive clutch of claim 2, wherein the first set of clutch members and the second set of clutch members alternate along the axis such that each clutch member of the first set adheres to two members of the second set when a voltage is applied.
4. The rotary electroadhesive clutch of claim 3, wherein the first set of clutch members and the second set of clutch members are compressed in an axial direction.
5. The rotary electroadhesive clutch of claim 3, wherein the first set of clutch members and the second set of clutch members are compressed using a bolt, rivet, or a fastener.
6. The rotary electroadhesive clutch of claim 2, wherein each of the first set of clutch members adheres to two members from the second set, each of which is directly adjacent to the member from the first set.
7. The rotary electroadhesive clutch of claim 6, wherein alignment of the clutch pairs is achieved using at least one of a shoulder or retaining ring located on the housing or shaft.
8. The rotary electroadhesive clutch of claim 1, wherein the first set of clutch members is constrained in rotation relative to a housing, and the second set of clutch members is constrained in rotation relative to a shaft.
9. The rotary electroadhesive clutch of claim 8, wherein the second set of clutch members transmits torque to the shaft and the first set of clutch members transmits torque to the housing through mating features that fit through holes in the clutch members, connections, or friction surfaces.
10. The rotary electroadhesive clutch of claim 1, wherein the structural frame comprises a material selected from the group consisting of fiberglass, carbon fiber, plastic, wood, paper, resin, cast epoxy, ceramic, a friction material, metal, or any combination of the foregoing.
11. The rotary electroadhesive clutch of claim 10, further comprising: an electrical connection from the voltage source to the clutch member, wherein the electrical connection is integrated into the structural frame or positioned on a surface of the structural frame.
12. The rotary electroadhesive clutch of claim 1, further comprising: an electrical connection between the voltage source and the clutch members, wherein the electrical connection is transmitted through an electrically conductive housing and/or an electrically conductive shaft in mechanical contact with a conductive surface of the clutch members.
13. The rotary electroadhesive clutch of claim 1, further comprising: an electrical connection, wherein the electrical connection between the voltage source and the clutch members is transmitted through a rigid or flexible wire, ribbon cable, flat flex cable, printed circuit board, slip ring, conductive spacers, or similar electrical connection.
14. The rotary electroadhesive clutch of claim 1, further comprising: an electrical connection, wherein the electrical connection between the voltage source and the clutch members is transmitted through a conductive brush, spring finger, spring-loaded pin, conductive washer, ring shim, battery contact, conductive gasket, conductive clip, swage, eyelet, non-conductive clamp holding two

conductive surfaces in contact, conductive pressure-sensitive adhesive, conductive epoxy, or soldering.

15. The rotary electroadhesive clutch of claim 1, further comprising: an electrical via, wherein the electrical via creates an electrical connection between a single insulating layer, multiple insulating layers, or physically separate electrically conductive surfaces.

16. A rotary electroadhesive clutch comprising: a first set of clutch members, wherein the first set of clutch members are electrically conductive; a second set of clutch members interposed between the first set of clutch members, wherein the second set of clutch members are electrically conductive; a dielectric material separating each clutch member of the first set of clutch members from an adjacent member of the second set of clutch members; a structural frame; an aperture in a perimeter of the structural frame; and an extension of the flexible film, wherein the extension passes from one side of the structural frame to the other through the aperture in order to enable an electrical connection to prevent contact between the electrically conductive portions of the first set of clutch members to electrically conductive portions of the second set of clutch members, wherein at least one of the first set of clutch members and the second set of clutch members comprises a flexible film, membrane, or sheet, in which the flexible film, membrane, or sheet is selectively or uniformly attached to the structural frame.

17. The rotary electroadhesive clutch of claim 16, wherein the electrical connection is integrated into the structural frame or positioned on a surface of the structural frame, and wherein the structural frame comprises a material selected from the group consisting of fiberglass, carbon fiber, plastic, wood, paper, resin, cast epoxy, ceramic, a friction material, metal, or any combination of the foregoing.

18. A rotary electroadhesive clutch comprising: a first set of clutch members, wherein the first set of clutch members are electrically conductive; a second set of clutch members interposed between the first set of clutch members, wherein the second set of clutch members are electrically conductive; a dielectric material separating each clutch member of the first set of clutch members from an adjacent member of the second set of clutch members; a structural frame; an opening, a slit, or an aperture in the structural frame; and an extension of the flexible film, wherein the extension passes from one side of the structural frame to the other in order to enable an electrical connection to prevent contact between the electrically conductive portions of the first set of clutch members to electrically conductive portions of the second set of clutch members, wherein at least one of the first set of clutch members and the second set of clutch members comprises a flexible film, membrane, or sheet, in which the flexible film, membrane, or sheet is selectively or uniformly attached to the structural frame.

19. The rotary electroadhesive clutch of claim 18, wherein the electrical connection is integrated into the structural frame or positioned on a surface of the structural frame, and wherein the structural frame comprises a material selected from the group consisting of fiberglass, carbon fiber, plastic, wood, paper, resin, cast epoxy, ceramic, a friction material, metal, or any combination of the foregoing.
